Sahaj Bhadja

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Summary

Automation software engineer with 2+ years designing, testing, validating, debugging, and sustaining **high-reliability software and data systems**; experienced in Linux, SQL, log analysis, integrations, installation, monitoring, root-cause analysis, and corrective action.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed and sustained **high-availability Python and SQL workflows** processing 5,000+ recurring monthly jobs with explicit states, validation, reconciliation, incremental loads, error handling, and 99%+ production reliability.
- Extracted and analyzed operational data across PostgreSQL and SQL Server, optimizing indexes and queries to reduce latency 35% while using structured diagnostics to isolate recurring integration failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Tested, validated, and debugged **concurrent Python processing services** across PDF, DOCX, image, and ZIP inputs, reproducing complex failures and implementing fallback handling, secure validation, and persistent results.
- Supported production workflows handling 10,000+ daily requests, analyzing error states and performance under peak load while shipping corrective fixes through GitHub Actions and reducing turnaround time 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Supported **integration, installation, and validation workflows** across Windows, Linux, and macOS VMs, automating nightly builds, executable deployment, end-to-end synchronization tests, coverage reports, and 20+ unit and integration tests.
- Performed root-cause analysis on API and ETL failures using Jenkins logs, SQL, AWS Glue, and S3, implementing corrective changes that improved response times 30% and pipeline efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Sustained **real-time production automation and data services** across eight Node.js components and 2M+ records, validating authentication, resource allocation, notifications, WebSocket synchronization, and Azure SQL behavior before release.
- Troubleshot high-priority system events using Sentry, Azure Monitor, and Kibana logs, coordinating tested Docker releases through Jenkins while reducing deployment time 50% and maintaining 95%+ build validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Collaborated with customers to identify requirements, map 12+ system interfaces, and document operational workflows for booking, pricing, payments, analytics, and real-time status updates across 50,000+ records.
- Developed reliable Flask and Firebase automation with validated webhooks, data synchronization, and transaction reconciliation, sustaining sub-200-millisecond responses while improving overall cross-platform compatibility 15% through targeted corrective changes.

Skills

Languages: Python, SQL, Bash, Java, JavaScript, TypeScript, C++
Systems: Unix/Linux, automation systems, high availability, installation, integration, validation, testing
Data: SQL Server, PostgreSQL, MySQL, Azure SQL, data extraction, analysis, modeling, indexing, query tuning
Operations: Troubleshooting, log analysis, root-cause analysis, corrective action, monitoring, escalation support
Delivery: Git, Jenkins, GitHub Actions, CI/CD, Docker, PyTest, Mockito, documentation, Agile collaboration

Projects

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*

- Built a **cross-platform SQL automation workspace** supporting Oracle, SQL Server, and 10+ systems with schema exploration, structured execution, data management, validation, Docker, Maven, and Linux-compatible debugging workflows.

CPU Scheduling Simulator Live | *C++, Algorithms, Validation*

- Implemented and validated **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case testing, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.

Big-Brain Live | *PostgreSQL, FAISS, Data Pipelines*

- Engineered a **data extraction and indexing pipeline** processing 50,000+ records with incremental updates, metadata persistence, validation, REST APIs, and performance tuning, reducing retrieval latency 40% through optimized transformations.